

SIH · IDEA SUBMISSION

AI-Powered Digital Evidence Management & Analysis System

GROUP MEMBERS

Snehal Ashok Supekar · Divya Somnath Salunke · Siddhi Ananda Thorat



Idea Title

01



THE CHALLENGE

Problem Statement

Managing digital evidence — images, videos, documents, emails, chat logs, etc. — manually is time-consuming and increases the risk of errors and data tampering.



OUR APPROACH

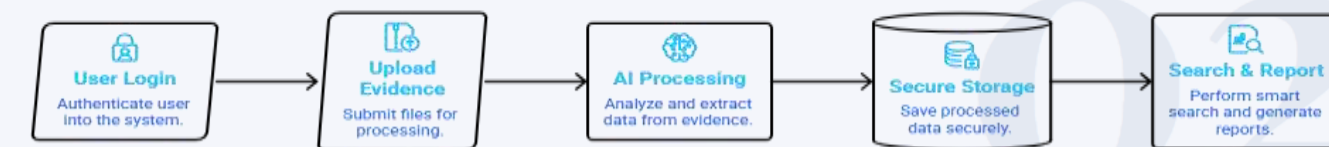
Proposed Solution

Develop a **secure web application** that uses AI to organize, analyze, and manage digital evidence efficiently.

Technical Approach

Technologies Used across the stack

AI Evidence Processing Workflow



Made with Napkin

01 • UI LAYER

Frontend

React.js
Tailwind CSS

02 • SERVER LAYER

Backend

Flask
Python

03 • DATA LAYER

Database

MongoDB
Document store

04 • INTELLIGENCE

AI

Python
OpenAI API

05 • LEARNING

Machine Learning

Scikit-learn
Classification & models

06 • TRUST

Security

bcrypt
SHA-256 Hashing

Feasibility & Viability

FEASIBILITY

Analysis of the Feasibility of the Idea

- Cost-effective using **open-source web technologies** and AI frameworks.
- Easily deployable as a **scalable and secure web application**.

CHALLENGES

Potential Challenges & Risks

- ! Protecting **sensitive digital evidence** from unauthorized access.
- ! Ensuring **accurate AI analysis** of different types of evidence.

STRATEGIES

Overcoming These Challenges

- ✓ Implement **secure authentication, encryption**, and access control.
- ✓ Regularly **train and validate AI models** using quality datasets.

Impact & Benefits

04

IMPACT

Potential Impact on the Target Audience

- 1 Enables **faster and more secure** digital evidence management.
- 2 Improves **investigation accuracy** with AI-powered analysis.

BENEFITS

Benefits of the Solution

SOCIAL

Supports **faster and more reliable** investigations.

ECONOMIC

Reduces **manual effort** and operational costs.

SECURITY

Protects **evidence integrity** and prevents unauthorized access.

Research & References

Academic sources supporting the proposed system

01

CONFERENCE PAPER · 2025

Enhancing Digital Investigation: The Role of Generative AI (ChatGPT) in Evidence Identification and Analysis in Digital Forensics

M. R. Hawa, M. Owda, and A. Y. Owda

02

JOURNAL ARTICLE · 2018

Digital Forensics: Review of Issues in Scientific Validation of Digital Evidence

Arshad, H., Jantan, A. B., & Abiodun, O. I. — *Journal of Information Processing Systems*, 14(2)

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LAW REVIEW · 2017

Authenticating Digital Evidence

Grimm, P. W., Capra, D. J., & Joseph, G. P. — *Baylor Law Review*, 69, 1